

Andhra Pradesh – Grade 10

2015 – Mathematics

1. Find L.C.M and H.C.F of 72 and 108 by the prime factorisation method.

Solution:

$$\begin{array}{r|l}
 2 & 72 \\
 \hline
 2 & 36 \\
 \hline
 2 & 18 \\
 \hline
 3 & 9 \\
 \hline
 3 & 3 \\
 \hline
 & 1
 \end{array}
 \qquad
 \begin{array}{r|l}
 2 & 108 \\
 \hline
 2 & 54 \\
 \hline
 3 & 27 \\
 \hline
 3 & 9 \\
 \hline
 3 & 3 \\
 \hline
 & 1
 \end{array}$$

$$72 = 2 \times 2 \times 2 \times 3 \times 3$$

$$108 = 2 \times 2 \times 3 \times 3 \times 3$$

$$\text{HCF} = 2 \times 2 \times 3 \times 3 = 36$$

$$\text{LCF} = 2 \times 2 \times 3 \times 3 \times 2 \times 3 = 216$$

2. If $V = \{a, e, i, o, u\}$ and $B = \{a, i, k, u\}$, then find $V - B$ and $B - V$. Are they equal?

Solution:

$$V - B = \{e, o\}$$

$$B - V = \{k\}$$

No; they are not equal

3. Find a quadratic polynomial, the sum and product of whose zeroes are $\frac{1}{4}$, -1 respectively.

Solution:

$$\text{sum} = \frac{1}{4} - 1 = \frac{-3}{4}$$

$$\text{product} = \frac{1}{4} \times -1 = \frac{-1}{4}$$

$$\text{equations: } x^2 - \left(\frac{-3}{4}\right)x - \frac{1}{4} = 0$$

$$4x^2 + 3x - 1 = 0$$

4. Find two numbers, whose sum is 27 and product is 182.

Solution:

$$\begin{array}{r|l}
 2 & 182 \\
 \hline
 13 & 91 \\
 \hline
 7 & 13 \\
 \hline
 & 1
 \end{array}$$

$\therefore 14$ and 13

5. For what value of 'k', the pair equations $3x + 4y + 2 = 0$ and $9x + 12y + k = 0$ represent co-incident lines.

Solution:

$$\text{Coincident lines} = \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$a_1 = 3 \quad b_1 = 4 \quad c_1 = 2$$

$$a_2 = 9 \quad b_2 = 12 \quad c_2 = k$$

$$\frac{3}{9} = \frac{4}{12} = \frac{2}{k}$$

$$k = \frac{2 \times 9}{3} = 6$$

6. In a flower bed, there are 23 rose plants in the first row, 21 in the second row, 19 in the third row, and so on. There are 5 rose plants in the last row. How many rows are there in the flower bed?

Solution:

23, 21, 19,, 5

$\therefore d = -2$ everywhere, it's A.P. $a_0 = 23 \quad d = -2$

$$= \frac{n}{2} (2a_0 + (n-1)d)$$

$$a_n = a_0 + (n-1)d$$

$$5 = 23 + (n-1) - 2$$

$$9 = n - 1$$

$$n = 10$$

$$= \frac{10}{2} (2 \times 23 + (10-1)(-2))$$

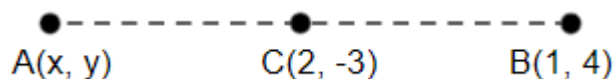
$$= 5 (46 - 18)$$

$$= 5 \times 28$$

$$= 140$$

7. Find the co-ordinates of a point A, where AB is the diameter of a circle, whose centre is (2, -3) and B is (1, 4).

Solution:



$$2 = \frac{x+1}{2}$$

$$4 - 1 = x$$

$$x = 3$$

$$-3 = \frac{y+4}{2}$$

$$-6 - 4 = y$$

$$y = -10$$

So, A (3, -10)

8. Find the area of the triangle, whose vertices are (-5, -1), (3, -5), (5, 2)

$$\begin{aligned}\text{Solution: Area} &= \frac{1}{2} \{x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)\} \\ &= \frac{1}{2} \{-5(-5 - 2) + 3(2 + 1) + 5(-1 + 5)\} \\ &= \frac{1}{2} \{35 + 9 + 20\} \\ &= 64 \times \frac{1}{2} = 32 \text{ cm}^2\end{aligned}$$

9. Expand $\log 15$.

Solution:

$$\begin{aligned}\log 15 \\ &= \log 5 \times 3 \\ &= \log 5 + \log 3\end{aligned}$$

10. Write roster and set builder form of "The set of all natural numbers, which divide 42".

Solution:

- (i) $x = \{1, 2, 3, 6, 7\}$
(ii) $x = \{x : x = 1, 2, 3, 6, 7\}$

11. If $p(t) = t^3 - 1$, find the values of $p(1)$, $p(-2)$.

Solution:

$$\begin{aligned}p(1) &= 1^3 - 1 = 0 \\ p(-2) &= (-2)^3 - 1 \\ &= -8 - 1 = -9\end{aligned}$$

12. Formulate a pair of linear equations in two variables "5 pencils and 7 pens together cost Rs. 50, whereas 7 pencils and 5 pens together cost Rs. 46".

Solution:

Let pencil = x

pen = y

$$5x + 7y = 50 \times 5$$

$$7x + 5y = 46 \times 7$$

$$25x + 35y = 250$$

$$49x + 35y = 322$$

$$\begin{array}{r} - \quad + \quad - \\ \hline \end{array}$$

$$24x = 72$$

$$x = \frac{72}{24} = 3$$

$$\text{Pencil} = 3$$

$$y = \frac{46 - 7x}{5} = \frac{46 - 7 \times 3}{5} = \frac{46 - 21}{5} = \frac{25}{5} = 5$$

$$\text{Pen} = 5$$

13. If $b^2 - 4ac \geq 0$, then write the roots of a quadratic equation $ax^2 + bx + c = 0$.

Solution: Real, equal and distinct.

14. Write the formula for the sum of first 'n' positive integers.

Solution:

$$\frac{n(n+1)}{2}$$

15. Prove that $\sqrt{5}$ is irrational by the method of Contradiction.

Solution:

Let $\sqrt{5} = \frac{a}{b}$ be rational

$$a = b\sqrt{5}$$

On L.H.S a is an rational number

On R.H.S $b\sqrt{5}$ is a irrational

Which contradicts the assumptions.

Hence $\sqrt{5}$ is irrational.

16. If $A = \{3, 6, 9, 12, 15, 18, 21\}$; $B = \{4, 8, 12, 16, 20\}$ $C = \{2, 4, 6, 8, 10, 12, 14, 16\}$; $D = \{5, 10, 15, 20\}$; find

(i) $A - B$,

(ii) $B - A$,

(iii) $C - A$,

(iv) $D - A$,

(v) $B - C$,

(vi) $B - D$,

(vii) $C - B$,

(viii) $D - B$

Solution:

(i) $A - B = \{3, 6, 9, 15, 18, 21\}$

- (ii) $B - A = \{4, 8, 16, 20\}$
- (iii) $C - A = \{2, 4, 8, 10, 14, 16\}$
- (iv) $D - A = \{5, 10, 20\}$
- (v) $B - C = \{20\}$
- (vi) $B - D = \{4, 8, 12, 16\}$
- (vii) $C - B = \{2, 6, 10, 14\}$
- (viii) $D - B = \{5, 10, 15\}$

17. Verify that 1, -1 and -3 are the zeroes of the cubic polynomial $x^3 + 3x^2 - x - 3$ and check the relationship between zeroes and the co-efficient.

Solution: To verify roots, put root inside equations

$$x^3 + 3x^2 - x - 3$$

$$@ \text{ Root } 1 \rightarrow 1^3 + 3(1)^2 - 1 - 3$$

$$1 + 3 - 1 - 3 = 0 \quad (\text{yes})$$

$$@ \text{ Root } (-1) \rightarrow -1^3 + 3(-1)^2 + 1 - 3 = -1 + 3 + 1 - 3 = 0 \quad (\text{yes})$$

$$@ \text{ Root } (-3) \rightarrow (-3)^3 + 3(-3)^2 - (-3) - 3 = -27 + 27 + 3 - 3 = 0 \quad (\text{yes})$$

$$\text{Sum of root from equation} = \frac{-b}{a} = \frac{-3}{1} = -3$$

$$\text{Form roots} = (1 - 1 - 3) = -3 \quad (\text{yes both matches})$$

$$\text{Product of roots} = \frac{-d}{a} = \frac{+3}{1} = 3$$

$$\text{From roots} = 1 \times -1 \times -3 = 3 \quad (\text{yes again matches})$$

$$\text{Product of roots } pq + qr + rp = \frac{c}{a} = \frac{-1}{1} = -1 \quad (\text{yes matches})$$

$$\text{From roots} = 1 \times -1 + -1 \times -3 + -3 \times 1 = -1 + 3 - 3 = -1 \quad (\text{yes matches})$$

Hence verified

18. Find the root of equation $2x^2 - x - 4 = 0$ by the method of completing the Square.

Solution:

$$2x^2 - x - 4 = 0$$

$$2x^2 - x = 4$$

$$x^2 - \frac{x}{2} = 2$$

$$x^2 - \frac{x}{2} + \left(\frac{1}{4}\right)^2 = 2 + \left(\frac{1}{4}\right)^2$$

$$\left(x - \frac{1}{4}\right)^2 = 2 + \frac{1}{16} = \frac{33}{16}$$

$$x - \frac{1}{4} = \frac{\pm\sqrt{33}}{4}$$

$$x = \frac{\pm\sqrt{33} + 1}{4}$$

19. Solve the equations: $\frac{5}{x-1} + \frac{1}{y-2} = 2$ and $\frac{6}{x-1} - \frac{3}{y-2} = 1$

Solution:

Let $\frac{1}{x-1} = A$ $\frac{1}{y-2} = B$

$$5A + B = 2 \times 3$$

$$6A - 3B = 1$$

$$15A + 3B = 6$$

$$6A - 3B = 1$$

$$21A = 7$$

$$A = \frac{1}{3}$$

$$\frac{1}{x-1} = \frac{1}{3}$$

$$3 = x - 1$$

$$x = 4$$

$$B = 2 - 5A$$

$$= 2 - 5 \times \frac{1}{3} = \frac{6-5}{3} = \frac{1}{3}$$

$$\frac{1}{y-2} = \frac{1}{3}$$

$$y = 3 + 2$$

$$y = 5$$

20. A fraction becomes $\frac{4}{5}$ if 1 is added to both numerator and denominator. If, however, 5 is subtracted from both numerator and denominator, the fraction becomes $\frac{1}{2}$. What is the fraction?

Solution:

Case 1:

$$\frac{x+1}{y+1} = \frac{4}{5} \quad \dots\dots(1)$$

$$\frac{x-5}{y-5} = \frac{1}{2} \quad \dots\dots(2)$$

$$= 5x + 5 = 4y + 4$$

$$5x - 4y = -1$$

$$= 2x - 10 = y - 5$$

$$2x - y = 5$$

Solve,

$$5x - 4y = -1$$

$$2x - y = 5 \times 4$$

$$8x - 4y = 20$$

$$\begin{array}{r} - \quad + \quad - \\ \hline -3x \quad = -21 \end{array}$$

$$x = 7$$

$$y = 2x - 5$$

$$= 9$$

$$\text{Fraction become} = \frac{7}{9}$$

21. The sum of the 4th and 8th terms of an A.P is 24 and the sum of the 6th and 10th terms is 44. Find the first three terms of the A.P.

Solution: $a_4 + a_8 = 24 \quad \dots\dots\dots(1)$

$$a_0 + 3d + a_0 + 7d = 24$$

$$2a_0 + 10d = 24$$

$$a_0 + 5d = 12 \quad \dots\dots\dots(1)$$

Also,

$$a_6 + a_{10} = 44$$

$$a_0 + 5d + a_0 + 9d = 44$$

$$2a_0 + 14d = 44$$

$$a_0 + 7d = 22 \quad \dots\dots\dots(2)$$

$$a_0 + 5d = 12$$

$$a_0 + 7d = 22$$

$$\begin{array}{r} - \quad - \quad - \\ \hline \end{array}$$

$$-2d = -10$$

$$d = 5$$

$$a_0 = 12 - 5d$$

$$= 12 - 5 \times 5$$

$$= -13$$

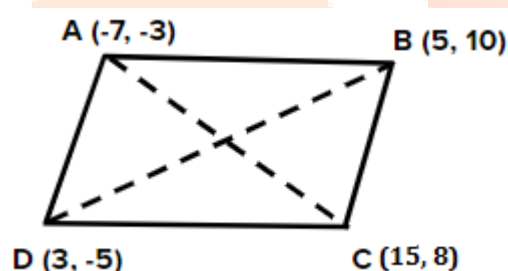
$$\text{First term} = a_0 = -13$$

$$\text{Second term} = a_0 + d = -13 + 5 = -8$$

$$\text{Third term} = a_0 + 2d = -13 + 10 = -3$$

22. Prove that the points $(-7, -3)$, $(5, 10)$, $(15, 8)$ and $(3, -5)$ taken in order are the vertices of a parallelogram.

Solution:



Let say ABCD are point of parallelogram

To prove: correct order of vertices, diagonals bisect each other in parallelogram.

Hence point O must be some

from A and C

$$C = \frac{-7+15}{2}, \frac{-3+8}{2}$$

$$= \left(4, \frac{5}{2}\right)$$

from B and D

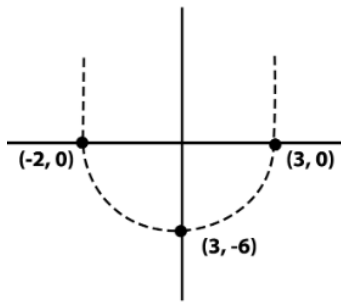
$$C = \frac{-5+3}{2}, \frac{10-5}{2}$$

$$= \left(4, \frac{5}{2}\right)$$

Hence, both are correct.

23. Draw the graph of $y = x^2 - x - 6$ and find zeroes. Justify the answer.

Solution:



@ x intercept $y = 0$

$$x^2 - x - 6 = 0$$

$$x^2 - 3x + 2x - 6 = 0$$

$$x(x - 3) + 2(x - 3) = 0$$

$$x = 3, x = -2$$

@ y intercept $x = 0$

$$y = 0 - 0 - 6 = -6$$

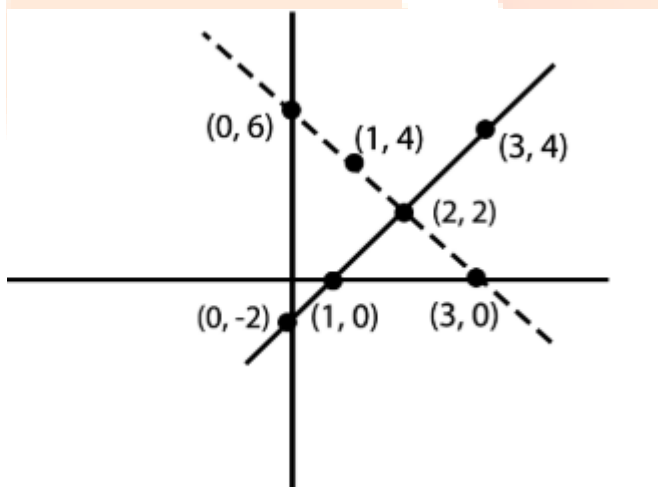
Point are = $(3, 0)$ $(-2, 0)$ $(0, -6)$

24. Solve the pair of linear equations graphically.

$$2x + y - 6 = 0$$

$$4x - 2y - 4 = 0$$

Solution:



$$y = 6 - 2x$$

x	0	1	2	3
y	6	4	2	0

$$y = \frac{4x - 4}{2} = 2x - 2$$

x	0	1	2	3
y	-2	0	2	4

Hence, from graph solution is $(2, 2)$